AC^{2}T-MARL: Causal Convolution and Curiosity for Offline-to-Online Multi-Agent Reinforcement Learning

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Abstract. Offline multi-agent reinforcement learning (MARL) learns cooperative policies from pre-collected trajectories without interacting with the environment during training. This setting is attractive when online exploration is expensive or unsafe, but it remains bottlenecked by two issues: imperfect supervision from suboptimal demonstrations and brittle adaptation once the policy reaches out-of-distribution (OOD) states that are weakly covered offline. We address this offline-to-online gap with AC^{2}T-MARL, a framework that couples a causal-convolutional sequence actor with curiosity-guided online refinement. Our approach introduces a sequence modeling actor named the Causal Decentralized Transformer (CDT), which replaces multi-head self-attention with causal depthwise-pointwise convolutions to inject stronger local temporal bias into decentralized trajectory modeling. Furthermore, we introduce an Intrinsic Curiosity Module for MARL (ICM) that supplies prediction-error intrinsic rewards during MAPPO-style fine-tuning to expand state coverage beyond the offline support. The resulting contribution is an offline-to-online co-design for Dec-POMDPs: a localized sequence backbone for offline policy learning together with a curiosity mechanism for crossing the offline-online coverage gap. Experiments on SMAC and Google Research Football show that AC^{2}T-MARL yields its clearest advantages on hard coordination-heavy tasks, while remaining competitive rather than uniformly dominant across all tasks. This work highlights the value of coupling localized offline sequence modeling with curiosity-guided adaptation for robust offline-to-online MARL.

Keywords: Offline-to-online MARL · Sequence modeling · Curiosity-driven exploration · Out-of-distribution generalization

1 Introduction

Multi-agent reinforcement learning (MARL) has become an important paradigm for cooperative decision making in decentralized partially observable Markov decision processes (Dec-POMDPs) [12, 13, 10]. It has been successfully applied to cooperative robotics, production systems, and emergency response, where agents must coordinate from local observations under non-stationary dynamics. In many real systems, however, large-scale online interaction is expensive, risky, or unavailable, which makes offline MARL especially attractive [2].

Existing offline MARL methods mainly follow two directions. The first direction extends temporal-difference learning with policy or value regularization, including BCQ, IQL, ICQ, and CQL variants [3, 7, 15, 8]. The second direction treats policy learning as sequence modeling and applies behavior imitation through architectures such as Decision Transformer, Decision ConvFormer, and MADT [1, 6, 11]. Sequence-modeling methods are appealing because they sidestep unstable offline value estimation and directly model temporally coherent action patterns.

For offline-to-online MARL, however, strong sequence modeling alone does not resolve the offline-to-online gap. Policies learned from suboptimal data can inherit local decision errors and fail once they encounter OOD states during deployment. Prior fine-tuning approaches alleviate this mismatch [11], but their effectiveness depends on two factors that remain under-addressed: whether the actor has the right inductive bias for decentralized trajectory modeling, and whether the online stage can explore beyond the support of the offline data without destabilizing coordination.

This paper addresses these two factors jointly. First, multi-agent trajectories contain strong local temporal structure: short-horizon action-observation dependencies often determine whether coordination succeeds under partial observability. Standard self-attention is expressive but does not explicitly favor this local structure. Second, offline-initialized policies often explore too conservatively during online refinement, so they adapt slowly when the environment shifts away from the offline data distribution. These observations motivate a unified design in which representation learning and online adaptation target the same offline-to-online failure mode.

We therefore propose AC^{2}T-MARL, an offline-to-online MARL framework that combines a causal-convolutional sequence model with intrinsic-curiosity-guided fine-tuning. We do not claim that causal convolution or curiosity is new in isolation; the technical novelty lies in adapting them as a coupled CTDE-compatible pipeline for offline-to-online Dec-POMDP control. Compared with MADT [11], our method changes the actor's token mixer to encode stronger local temporal bias and augments online adaptation with explicit exploration pressure. Compared with Decision ConvFormer [6], our focus is decentralized multi-agent coordination and the offline-to-online transition, not single-agent decision making alone. Accordingly, AC^{2}T-MARL combines a localized sequence backbone for offline policy learning with curiosity-guided online refinement for crossing the offline-online coverage gap.

As shown in Fig. 1, AC^{2}T-MARL contains two complementary modules. CDT replaces multi-head self-attention with a causal convolutional token mixer so that the actor can model local temporal dependencies more directly while preserving the residual and feed-forward structure of the Transformer backbone. ICM augments the online fine-tuning stage with intrinsic rewards derived from transition prediction errors, encouraging agents to visit poorly covered states and adapt more effectively to OOD situations.

The main contributions of this paper are as follows.

- We formulate offline-to-online MARL as a co-design problem between local sequence representation and online coverage expansion, highlighting why decentralized imitation alone remains brittle under partial observability and OOD adaptation.
- We propose AC^{2}T-MARL, whose CDT introduces causal local token mixing for decentralized history modeling and whose ICM augments CTDE fine-tuning with explicit exploration pressure while preserving decentralized execution.
- We evaluate the resulting design on SMAC and Google Research Football, using SMAC as the main benchmark suite and Football as a second cooperative domain; across both settings, the combined pipeline is most beneficial on coordination-heavy tasks where offline coverage is weakest.

2 Related Work

Offline MARL methods are often grouped into value-regularized temporal-difference approaches and sequence-modeling approaches [2]. BCQ-MA, ICQ-MA, and CQL-style methods constrain policy improvement around the dataset support through conservative value learning or implicit policy regularization [3, 7, 15, 8]. These methods are effective when value estimation remains reliable, but they can still be brittle under severe distributional shift and difficult long-horizon coordination.

Sequence-modeling methods instead learn policies by directly modeling action trajectories. Decision Transformer reformulates reinforcement learning as autoregressive sequence prediction [1], Decision ConvFormer replaces attention with convolutional token mixing for single-agent decision making [6], and MADT extends this line to offline multi-agent control [11]. Our work is closest to MADT in overall training protocol, but differs in two ways that are central to the offline-to-online setting studied here: we replace self-attention with a causal convolutional mixer tailored to localized Dec-POMDP histories, and we couple the resulting offline policy with explicit curiosity-driven adaptation during the online stage.

Offline-to-online fine-tuning adds a second layer of difficulty because the pretrained policy must improve without drifting too far from the coordination patterns encoded in the dataset. Existing MARL fine-tuning pipelines typically inherit the offline actor and then rely on PPO-style updates for adaptation [11, 16]. In contrast, we explicitly treat representation bias and online exploration as

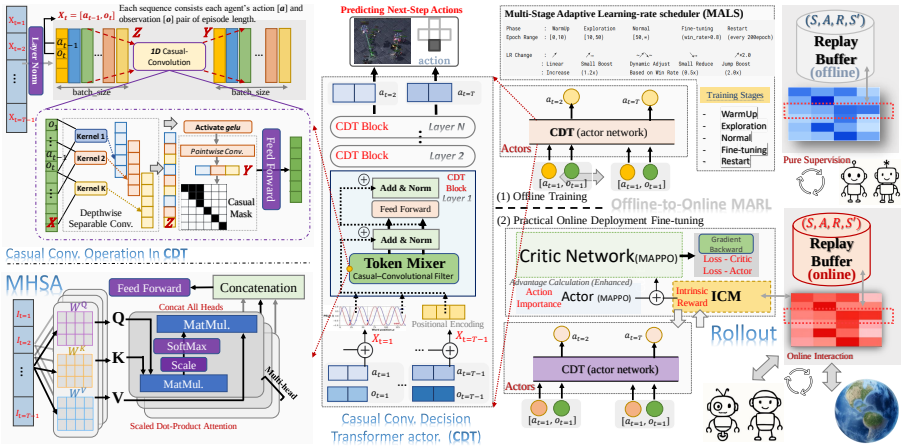


Fig. 1. Overview of AC^{2}T-MARL. The framework performs offline pre-training with the proposed Causal Decentralized Transformer (CDT) and then conducts online fine-tuning with the Intrinsic Curiosity Module (ICM). The left side illustrates the replacement of the standard Transformer token mixer by causal convolution, while the right side summarizes the offline-to-online pipeline.

a coupled design problem: the offline actor should expose short-horizon coordination cues clearly enough for adaptation to start from a strong local policy, while the online stage should broaden coverage without discarding that initialization.

Our online refinement stage is also related to curiosity-driven exploration. The difference is positional rather than purely architectural: ICM is used here to support adaptation of an offline-pretrained decentralized sequence policy during online refinement, rather than as a generic exploration bonus for randomly initialized MARL. This positioning is important because the policy begins online learning from a conservative imitation prior, so the exploration module must widen state coverage without breaking coordination.

3 Method

3.1 Overview

AC^{2}T-MARL follows a two-stage offline-to-online learning pipeline. In the offline stage, we pre-train a decentralized sequence actor on demonstration trajectories. In the online stage, we fine-tune the pre-trained actor with MAPPO-style updates while augmenting the reward signal with intrinsic curiosity. The resulting design is modular but targeted: CDT improves the trajectory representation used to initialize the policy, and ICM improves exploration once the policy moves beyond the support of the offline dataset. The pipeline is: train CDT by behavior imitation, initialize online MAPPO with the learned actor, compute extrinsic plus ICM intrinsic rewards from online rollouts, and update the decentralized actor together with the centralized critic under the CTDE protocol. This division is important for the offline-to-online setting because the two modules

address different failure points: CDT reduces the imitation-stage mismatch from local histories, whereas ICM encourages coverage expansion after the initialized policy reaches states that are weakly represented offline.

3.2 Causal Decentralized Transformer

For each agent i, the CDT actor consumes a local history prefix

$$\tau_t^i = (o_1^i, a_1^i, \ldots, o_{t-1}^i, a_{t-1}^i, o_t^i), \tag{1}$$

where o_t^i denotes the local observation. The joint policy is factorized in the usual decentralized form

$$\pi_\theta(\mathbf{a}_t \mid \boldsymbol{\tau}_t) = \prod_{i=1}^{N} \pi_\theta^i(a_t^i \mid \tau_t^i), \tag{2}$$

and each actor predicts the next action a_t^i from its own local history. In implementation, each history is tokenized as an interleaved sequence of observation and previous-action embeddings,

$$\mathbf{X}_t^i = [E_o(o_1^i), E_a(a_1^i), \ldots, E_o(o_t^i)] + E_{\text{time}}, \tag{3}$$

where E_o and E_a denote learned embedding layers and E_{time} denotes timestep encoding. For the homogeneous agents considered in SMAC, the same actor parameters are shared across agents; for Google Research Football, the same shared actor is used together with a learned role embedding that marks the controlled player identity. Shorter histories are left-padded within a batch and padded positions are excluded from the imitation loss. Let $\mathbf{X} \in \mathbb{R}^{B \times T_\tau \times d}$ denote the embedded token sequence, where B is the batch size, T_τ is the token length, and d is the embedding dimension. The hidden state aligned with the current observation token is

$$\mathbf{h}_t^i = f_\theta(\mathbf{X}_t^i)_{|\tau_t^i|}, \tag{4}$$

where $|\tau_t^i| = 2t - 1$ because the current observation occupies the final odd token position in the interleaved history. The policy head maps this token to the next-action distribution,

$$\pi_\theta^i(a_t^i \mid \tau_t^i) = \text{Softmax}(W_a \mathbf{h}_t^i + b_a). \tag{5}$$

If a sampled trajectory window contains H decision steps, the offline objective is the masked autoregressive behavior-cloning loss

$$\mathcal{L}_{\text{off}} = -\sum_{i=1}^{N} \sum_{t=1}^{H} m_t^i \log \pi_\theta^i(a_t^i \mid \tau_t^i), \tag{6}$$

where $m_t^i = 0$ on padded positions and $m_t^i = 1$ otherwise. Unlike return-conditioned single-agent DT variants, CDT uses only the local-history interface above, which allows the same actor definition to be reused unchanged during

online PPO fine-tuning. Using this interface in both stages yields a consistent policy formulation for offline pre-training and online adaptation.

The key change in CDT is the token mixer. Instead of standard multi-head self-attention, we apply a causal convolutional module that preserves temporal order while emphasizing short-range temporal interactions. The transformation contains four steps:

1. The input tensor is layer-normalized and transposed from (B, T_τ, d) to (B, d, T_τ) so that one-dimensional convolution can operate along time.
2. A left-padded depthwise causal convolution extracts channel-wise local temporal patterns:

$$\mathbf{Z}_{\mathrm{dw}} = \mathrm{DWConv1D}\big(\mathrm{Pad}_{K-1}(\mathbf{X}^\top); \mathbf{W}_{\mathrm{dw}}\big)\,, \tag{7}$$

   where K is a fixed kernel width shared across maps, the dilation is 1, and the zero padding is applied only on the left. With stride 1, each output at token index u depends only on tokens from indices $\leq u$, which enforces token-level causality on the interleaved observation-action sequence.
3. A pointwise convolution mixes information across channels:

$$\mathbf{Z}_{\mathrm{pw}} = \mathrm{PWConv1D}(\mathbf{Z}_{\mathrm{dw}}; \mathbf{W}_{\mathrm{pw}})\,. \tag{8}$$

4. The output is transposed back to (B, T, d) and projected to the original feature dimension:

$$\mathbf{Y} = \mathrm{Proj}\big(\mathbf{Z}_{\mathrm{pw}}^\top\big)\,. \tag{9}$$

   For brevity, the token-mixing block can be written as

$$\mathbf{Y} = \mathrm{Proj}(\mathrm{CausalConv}(\mathrm{LayerNorm}(\mathbf{X})))\,. \tag{10}$$

The output then passes through the same residual connection and feed-forward sublayers as a standard Transformer block. Because the action head always reads the hidden state aligned with the current observation token, token-level causality is sufficient to guarantee that the prediction of a_t^i depends only on $(o_1^i, a_1^i, \ldots, o_t^i)$ and never on future actions or observations. This replacement preserves the overall architecture while injecting stronger local temporal bias for decentralized trajectory modeling under partial observability.

3.3 Intrinsic Curiosity Module for MARL

During online fine-tuning, the actor must adapt to state-action configurations that are weakly represented in the offline data. To encourage useful exploration, we adopt an Intrinsic Curiosity Module (ICM) that produces intrinsic rewards from transition prediction errors.

As illustrated in Fig. 2, ICM contains feature encoders, an inverse model, and a forward model. Following the centralized-training-decentralized-execution protocol, ICM uses the centralized state s_t during training, while the policy itself remains conditioned only on decentralized history inputs, namely local histories

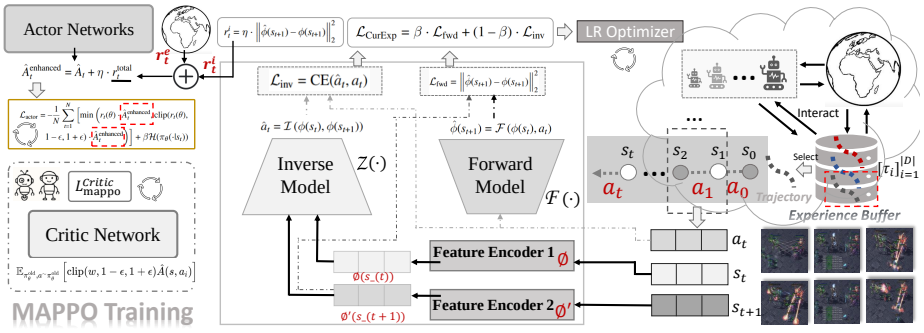


Fig. 2. Architecture of the Intrinsic Curiosity Module (ICM). The module uses current state, joint action, and next state to learn forward and inverse dynamics and produces an intrinsic reward that is added to the extrinsic reward during online fine-tuning.

τ_t^i together with optional fixed role embeddings. Thus, the deployed actor does not require global state or the curiosity model, but training still assumes access to centralized information. This assumption matches the SMAC and Google Research Football setups used in our experiments; in settings without centralized training information, the curiosity module would need a learned state surrogate from local histories. The forward model predicts the feature representation of the next state from the current state and the joint action:

$$\hat{\phi}(s_{t+1}) = \mathcal{F}(\phi(s_t), \mathbf{a}_t)\,. \tag{11}$$

The inverse model predicts the executed action for each agent from the current and next-state features:

$$\hat{\mathbf{a}}_t = \{\hat{a}_t^1, \ldots, \hat{a}_t^N\} = \mathcal{I}(\phi(s_t), \phi(s_{t+1}))\,. \tag{12}$$

The intrinsic reward is defined as the scaled forward-prediction error:

$$r_t^{\text{int}} = \eta \left\| \hat{\phi}(s_{t+1}) - \phi(s_{t+1}) \right\|_2^2, \tag{13}$$

where η controls the strength of curiosity. The inverse and forward models are optimized by

$$\mathcal{L}_{\text{inv}} = \sum_{i=1}^{N} \text{CE}\big(\hat{a}_t^i, a_t^i\big)\,, \tag{14}$$

$$\mathcal{L}_{\text{fwd}} = \left\| \hat{\phi}(s_{t+1}) - \phi(s_{t+1}) \right\|_2^2, \tag{15}$$

and the total ICM loss is

$$\mathcal{L}_{\text{ICM}} = \beta \mathcal{L}_{\text{fwd}} + (1-\beta)\mathcal{L}_{\text{inv}}, \tag{16}$$

where β balances the two objectives.

During online fine-tuning, the total reward is

$$r_t^{\text{total}} = r_t^{\text{ext}} + r_t^{\text{int}}, \tag{17}$$

where r_t^{ext} is the environmental reward. We then compute the advantage from the total reward with standard GAE:

$$\hat{A}_t^{\text{enh}} = \text{GAE}(r_t^{\text{total}}, V_\psi(s_t)), \tag{18}$$

where the critic $V_\psi(s_t)$ estimates the return induced by the total reward in Eq. (17). The corresponding critic objective is

$$\mathcal{L}_{\text{critic}} = \frac{1}{M} \sum_{t=1}^{M} \left(V_\psi(s_t) - \hat{G}_t^{\text{total}} \right)^2, \tag{19}$$

where $\hat{G}_t^{\text{total}}$ is the bootstrapped centralized team-return target constructed from truncated online rollouts:

$$\hat{G}_t^{\text{total}} = V_\psi(s_t) + \sum_{l=0}^{L-1} (\gamma \lambda_{\text{GAE}})^l \delta_{t+l}, \quad \delta_t = r_t^{\text{total}} + \gamma V_\psi(s_{t+1}) - V_\psi(s_t). \tag{20}$$

Here λ_{GAE} is the standard GAE trace-decay parameter and should not be confused with the loss weights λ_V, λ_I, and λ_H below. We optimize the decentralized actor with a PPO-style clipped objective:

$$\begin{aligned} \mathcal{L}_{\text{actor}} = &- \frac{1}{M} \sum_{t=1}^{M} \min \left(\rho_t(\theta) \hat{A}_t^{\text{enh}}, \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_t^{\text{enh}} \right) \\ &- \lambda_H \sum_{i=1}^{N} \mathcal{H}(\pi_\theta^i(\cdot \mid \tau_t^i)), \end{aligned} \tag{21}$$

where M is the minibatch size and

$$\rho_t(\theta) = \prod_{i=1}^{N} \frac{\pi_\theta^i(a_t^i \mid \tau_t^i)}{\pi_{\theta_{\text{old}}}^i(a_t^i \mid \tau_t^i)}. \tag{22}$$

The online stage jointly minimizes

$$\mathcal{L}_{\text{online}} = \mathcal{L}_{\text{actor}} + \lambda_V \mathcal{L}_{\text{critic}} + \lambda_I \mathcal{L}_{\text{ICM}}. \tag{23}$$

Within each PPO epoch, the bootstrapped targets $\hat{G}_t^{\text{total}}$ and advantages $\hat{A}_t^{\text{enh}}$ are computed from truncated rollouts using a detached critic snapshot and are treated as constants during optimization; terminal transitions use zero bootstrap and advantages are normalized per minibatch. This design encourages policies that are both reward-seeking and sufficiently exploratory when the current policy drifts outside the support of the offline dataset.

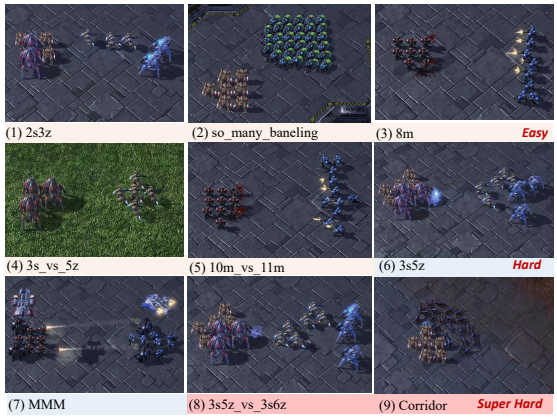


Fig. 3. Representative SMAC scenarios used in our evaluation, covering maps with increasing coordination difficulty.



Fig. 4. Google Research Football scenarios used in our evaluation, spanning full-game stochastic play and compact academy tasks with different coordination demands.

4 Experiments

4.1 Experimental Setup

We evaluate AC^{2}T-MARL on two cooperative MARL benchmarks: the StarCraft Multi-Agent Challenge (SMAC) [14] and Google Research Football [9]. SMAC contains tasks with diverse unit compositions, tactical constraints, and coordination difficulty. Figure 3 shows representative SMAC scenarios used in our experiments.

For Google Research Football, we use four Football Academy scenarios with progressively harder cooperative control demands: 11-vs-11 easy stochastic, 11-vs-11 hard stochastic, 5-vs-5, and 3-vs-1 with keeper. Figure 4 summarizes these scenarios.

We compare against several strong offline MARL baselines spanning different methodological families: BC, BCQ-MA, ICQ-MA, CQL-MA, and MADT [5, 3, 4, 15, 8, 11]. For both benchmarks, the offline dataset is constructed from MAPPO-generated trajectories and processed following the MADT protocol [16, 11]. For SMAC, each map uses approximately 2×10^4 trajectories, corresponding to roughly 2.6–3.1 million transitions depending on episode length; for Google Research Football, each scenario uses approximately 1.2×10^4 trajectories, or

about 1.3–1.6 million transitions. This controlled source makes methods comparable and isolates the effect of the model and online adaptation strategy, but it is less heterogeneous and noisy than real offline logs. The present evaluation should therefore be read as a controlled benchmark study rather than a complete test over expert, medium, random, and mixed-quality datasets.

All methods use the same offline data source, the same online fine-tuning budget, and the same evaluation cadence. Specifically, we fine-tune for 1.0 million environment steps on SMAC and 0.6 million steps on Google Research Football, evaluate every 10 000 steps, and report means over three random seeds. The 12-map SMAC panel omits confidence bands for readability. In our implementation, the CDT actor uses two sequence blocks with feature dimension 128, maximum token length 400, and convolution kernel width $K = 5$; this single kernel width is used across tasks without map-specific tuning. Since K controls the temporal locality bias of CDT, wider or dilated kernels could be beneficial when useful coordination cues span longer histories. The online stage uses PPO clipping $\epsilon = 0.2$, rollout horizon $L = 128$, intrinsic-reward coefficient $\eta = 0.01$, ICM mixing coefficient $\beta = 0.2$, critic weight $\lambda_V = 1.0$, curiosity weight $\lambda_I = 0.1$, and entropy coefficient $\lambda_H = 0.01$. For Google Research Football, the decentralized actor shares parameters across controllable players and appends the learned role embedding described in Sec. 3.2, while the centralized critic and ICM use the environment-provided global state under the same CTDE protocol. We report online curves, top-20 mean return, and final-window mean return where space permits. Top-20 mean return averages the 20 best evaluation points and is therefore peak-oriented; final-window mean return averages the last 10 evaluations and better reflects late-stage stability.

4.2 Main Results

Figure 5 reports seed-averaged online fine-tuning performance on 12 SMAC maps. AC^{2}T-MARL does not dominate every task, but it gives clear improvements on several difficult maps such as `corridor` and `3s5z vs. 3s6z`, where the coverage gap of offline data is more severe and online adaptation is most demanding. On maps such as `MMM`, `3s5z`, and `3s vs. 4z`, MADT remains slightly higher under the top-20 metric. We therefore interpret the results as evidence for stronger hard-task adaptation rather than uniform superiority across all maps.

Table 1 summarizes the top-20 mean return during online fine-tuning as a peak-oriented summary, not as the sole stability metric. Under this secondary metric, AC^{2}T-MARL achieves 8 outright wins, 1 tie, and 3 runner-up finishes across the 12 maps, with an average gain of +0.27 over MADT. The largest margins appear on complex coordination-heavy tasks, including `corridor`, `2s3z`, and `10m vs. 11m`. We read these results together with the learning curves and the final-window spot check in Table 2.

To further test cross-domain robustness, Table 3 reports online performance on Google Research Football. AC^{2}T-MARL achieves the best result on three of the four scenarios and remains close to MADT on the remaining one. The largest margin appears on the hard 11-vs-11 stochastic setting, where exploration under

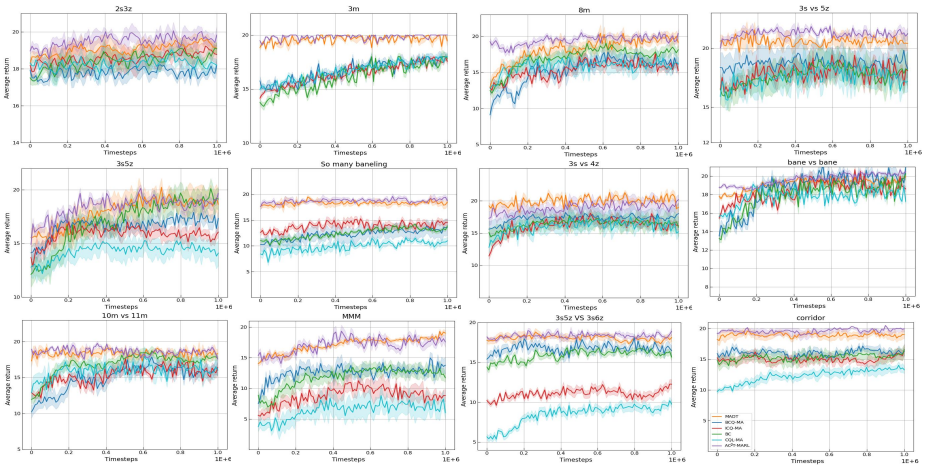


Fig. 5. Online fine-tuning results on 12 SMAC maps. AC^{2}T-MARL attains higher returns on many maps, with especially clear gains on harder scenarios that require broader online adaptation.

sparse rewards and long-horizon coordination is most demanding. Although the Football suite is smaller than the SMAC benchmark set, it points in the same direction.

Figure 6 reports pure offline performance before online fine-tuning. The AC^{2}T-MARL actor matches or surpasses the baselines on most maps, particularly on difficult scenarios where distributional shift and long-horizon coordination are more pronounced. This result is important because it shows that the gain of AC^{2}T-MARL does not come only from curiosity-driven exploration; the localized sequence actor used to initialize online adaptation is already competitive before fine-tuning begins.

Figure 7 studies the effect of curiosity-driven exploration. Adding ICM improves online fine-tuning across the reported tasks, with the clearest benefit on harder maps where exploration beyond the offline dataset is essential. In other words, the curiosity term primarily changes coverage and adaptation speed, whereas the CDT backbone primarily changes the quality of the starting policy that online refinement operates on.

4.3 Ablation Studies

We ablate the two main components of AC^{2}T-MARL to clarify their roles in offline-to-online adaptation. Figure 7 shows that adding ICM consistently improves adaptation speed and late-stage return, with the clearest gains on harder maps where exploration beyond the offline dataset is most important. Figure 8 compares the causal-convolutional token mixer with self-attention and shows that localized token mixing leads to more favorable adaptation dynamics on difficult maps. Taken together, these results suggest that CDT provides a stronger

Table 1. Top-20 mean return during online fine-tuning. Best and second-best values are highlighted in bold and underlined, respectively.

Map	AC^{2}T-MARL	MADT	BCQ-MA	ICQ-MA	BC	CQL-MA
3m	**20.00**	**20.00**	<u>17.86</u>	17.69	17.51	17.83
8m	**20.01**	<u>19.89</u>	17.21	16.89	18.30	16.77
2s3z	**19.66**	19.07	17.95	18.22	17.30	<u>19.27</u>
MMM	<u>18.33</u>	**18.40**	14.32	10.42	13.39	7.65
3s5z	<u>18.87</u>	**18.99**	15.46	15.06	14.30	10.94
3s vs. 5z	**21.60**	<u>21.16</u>	19.31	18.83	18.47	18.47
3s vs. 4z	<u>19.86</u>	**20.69**	17.91	17.50	17.07	17.14
10m vs. 11m	**19.27**	<u>18.76</u>	17.56	16.82	18.54	16.98
bane vs. bane	**20.34**	19.91	<u>20.24</u>	19.79	19.38	18.86
so_many baneling	**19.22**	<u>18.78</u>	13.28	14.84	13.67	11.05
3s5z vs. 3s6z	**18.46**	<u>18.06</u>	17.09	11.86	15.82	5.42
corridor	**19.61**	<u>18.31</u>	15.71	15.06	13.03	9.68

Table 2. Final-window mean return over the last 10 evaluations on representative hard tasks. This metric is less peak-sensitive than top-20 return and serves as a compact stability-oriented spot check.

Benchmark	Task	AC^{2}T-MARL	MADT
SMAC	corridor	**18.94**	17.85
SMAC	3s5z vs. 3s6z	**17.81**	17.39
GFootball	11-vs-11 hard stochastic	**0.84**	0.71
GFootball	5-vs-5	**1.74**	1.58

coordination-aware starting point, while ICM becomes most valuable when online fine-tuning must push the policy beyond the support of the offline data.

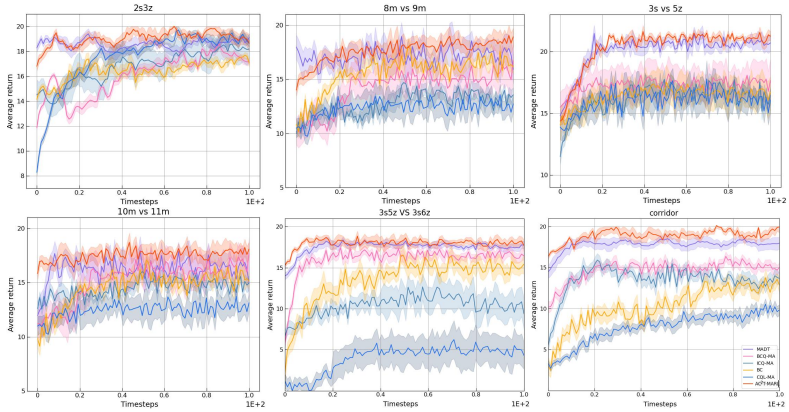


Fig. 6. Pure offline performance of AC^{2}T-MARL and baseline methods. The full offline actor yields a competitive starting point for later online adaptation.

Table 3. Google Research Football results measured by average episodic return during online fine-tuning. Best and second-best values are highlighted in bold and underlined, respectively.

Scenario	AC^{2}T-MARL	MADT	BCQ-MA	ICQ-MA	BC	CQL-MA
Academy_11_vs_11 _easy_stochastic	**1.43**	<u>1.31</u>	1.08	1.02	0.94	0.99
Academy_11_vs_11 _hard_stochastic	**0.96**	<u>0.77</u>	0.61	0.58	0.46	0.52
Academy_5_vs_5	**1.88**	<u>1.69</u>	1.37	1.29	1.21	1.25
Academy_3_vs_1 _with_keeper	<u>1.61</u>	**1.64**	1.32	1.27	1.18	1.21

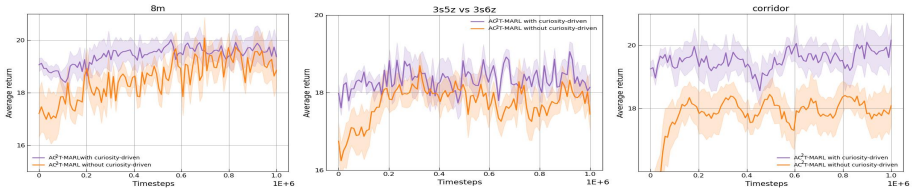


Fig. 7. Ablation of the Intrinsic Curiosity Module. Curiosity-guided fine-tuning improves adaptation speed and late-stage return on the three reported maps.

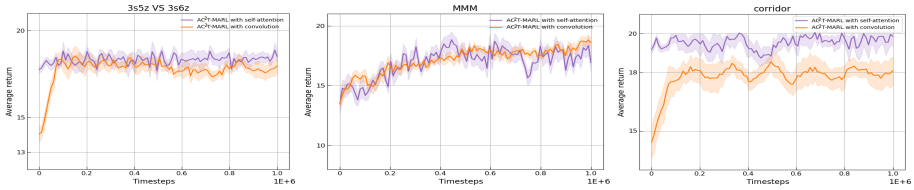


Fig. 8. Architectural ablation between self-attention and causal convolution in the actor during online fine-tuning. The curves indicate that token-mixer choice materially changes adaptation behavior on difficult maps.

4.4 Case Studies

Figure 9 provides a qualitative comparison between AC^{2}T-MARL and MADT on the super-hard map `3s5z vs. 3s6z`. The snapshots should be interpreted as illustrative rather than as an independent quantitative metric. AC^{2}T-MARL tends to preserve tighter formations and more consistent focus fire in the late stage, whereas MADT exhibits earlier fragmentation under pressure. These behaviors are consistent with the curve-based evidence that the proposed pipeline is most useful on hard coordination-heavy tasks.



Fig. 9. Real-time battle pieces in the '3s5z vs.3s6z'.

5 Conclusion

We presented AC^{2}T-MARL, an offline-to-online MARL framework that combines a causal-convolutional decentralized actor with curiosity-driven online fine-tuning. Overall, the results suggest that offline representation quality and online exploration should be designed jointly in decentralized MARL: localized sequence modeling can provide a competitive offline starting point, while ICM helps the initialized policy cross the offline-online coverage gap during online interaction. Experiments on SMAC provide the main evidence that this combination improves hard-map adaptation relative to strong offline MARL baselines, while a smaller Google Research Football study points in the same direction on a second cooperative domain. The current formulation remains tied to CTDE because curiosity and value learning rely on centralized state information during training. Its evaluation also uses homogeneous MAPPO-generated datasets and a fixed convolution width $K = 5$. In future work, we will study fully decentralized curiosity surrogates, mixed-quality real-world offline data, heterogeneous-agent settings, and broader kernel-sensitivity analysis.

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